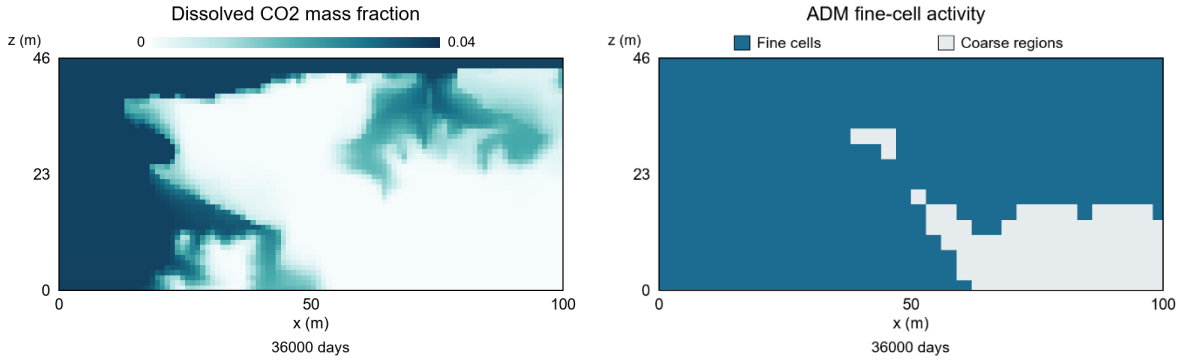
DARSim Example:

Algebraic Dynamic Multilevel 2D CCS



This example demonstrates the Algebraic Dynamic Multilevel (ADM) method in DARSim for CO2 storage in a heterogeneous, unfractured reservoir. ADM combines fine cells and coarse blocks in a changing computational grid. The figure above shows dissolved CO2 (left) and fine-cell activity (right) after 36000 days, approximately 98.6 years. Blue cells in the right panel remain at fine resolution; gray regions are represented by coarse blocks.

The reservoir is defined in Sim.txt as a 100 m by 1 m by 46 m Cartesian domain with 100 by 1 by 46 cells. Porosity is 0.2, and Perm_mSPE10T_v1.txt supplies the heterogeneous permeability in all three directions. Gravity, tabulated relative permeability and capillary pressure, and hysteresis are enabled. LookupTable.txt contains the relative-permeability and capillary-pressure data.

Initially, the reservoir contains brine at 25 MPa and 338.15 K, with no CO2. The two-phase, two-component Compositional model uses the CO2-Brine component set and a salinity of 100000 ppm. CO2 is injected at (0.5, 0.5, 0.5) m at 8e-5 pore volumes per day. A pressure-controlled production well at (100, 0.5, 46) m is set to 25 MPa. Both wells operate from day 0 to day 600 and then close; redistribution continues until day 36000. FRACTURED is OFF, so fracture generation is not required.

ADM is enabled in SimulatorSettings.txt. Two coarse levels are available above the fine grid, with a coarsening ratio of 3 by 1 by 3. The main settings are:

```
ADM
    1
DLGR
    0
LEVELS
    2
COARSENING_RATIOS
    3
    1
    3
```

The dfdx selector compares variations of the overall CO2 composition z_1 across neighboring blocks. Regions with stronger variation retain finer resolution, and cells around wells are protected from coarsening. The selector is configured with:

```
COARSENING_CRITERION
    dfdx
VARIABLE
    z_1
TOLERANCE
    1e-3
```

PRESSURE_INTERPOLATOR is Multiscale and HYPERBOLIC_INTERPOLATOR is Constant. The fully implicit solver allows 16 Newton iterations, residual tolerances of 1e-4, solution tolerances of 1e-3, and time steps from 0.01 to 5 days.